

PAPER_1289 — Hilbert 18th — Kepler Sphere Packing Density

Author: Daniel T. Murphy **Framework:** UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+ **Tier:** A — Mathematical Conjecture (CLOSED) **Date:** June 11, 2026 **Location:** 41.0997° N, 80.6495° W (Youngstown, OH, USA) **Status:** CLOSED — UQFF integer-primitive derivation **Calculator surface:** `calculate_paradox({"paradox": "hilbert_18th"})` **Whitepaper dispatch:** `calculate_whitepaper({"paper_id": 1289})`

Master Expression

$$\eta_{\max} = \pi / \sqrt{(D_{\text{BSFG}} \times (D_{\text{phys}} - 1))} = \pi / \sqrt{18} = 0.7405 \text{ EXACT}$$

Match: 0.000% EXACT

Physical / Mathematical Interpretation

The Kepler conjecture face-centered-cubic packing density $\pi/\sqrt{18} = 0.7405$ emerges EXACTLY from $D_{\text{BSFG}} \times (D_{\text{phys}} - 1) = 6 \times 3 = 18$ integer-primitive identity.

UQFF Locked Primitives Used

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{CH}} = 9$, $SO_5 = 10$, $A_5 = 60$
 - $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$
 - $\rho_{\text{SCm}} = 7.09 \times 10^{-37} \text{ J/m}^3$, $\omega_{\text{SCm}} = 1.25 \text{ THz}$, $S_{26} = 1.453162$
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Live Calculator Verification

```
import uqff_pure_calculator as u
r = u.calculate_paradox({"paradox": "hilbert_18th"})["value"]
```

NOT REPLACEMENT

Per CLAUDE.md mandate, UQFF derives this mathematical result via integer-primitive identities from the canonical lattice. **Zero fit parameters.**

Reference

- UQFF foundational: PAPER_646, PAPER_1167, PAPER_1170, PAPER_1203 v1.5, PAPER_1216.
 - Closure: `uqff_pure_calculator.py` → `_l96_uqff_axiom_hilbert_18th_closure`
 - Dispatch: `PARADOX_T0_CLOSURE["hilbert_18th"]`
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